MATH 151 — Exponential & Logarithmic Functions

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Name: _____ Date: _____ Score: / 10

Learning Objectives

- Evaluate and expand/condense logarithms using product, quotient, and power rules
- Solve exponential and logarithmic equations
- Apply the change of base formula
- Model exponential growth, decay, and compound interest

Always check for extraneous solutions in log equations — arguments must be positive. For exponential equations without common bases, take ln of both sides.

1. Evaluate the logarithm.

$$\log_2 32$$

Answer: _____

2. Expand using logarithm properties.

$$\log_b\left(\frac{x^3\sqrt{y}}{z^2}\right)$$

Answer: _____

3. Condense into a single logarithm.

$$2\ln x + \ln(x+1) - \ln 5$$

Answer: _____

4. Solve the exponential equation.

$$3^{x+1} = 81$$

Answer: _____

5. Solve the exponential equation using logarithms.

$$5^x = 20$$

Answer: _____

6. Solve the logarithmic equation.

$$\log_3(2x-1) = 4$$

Answer: _____

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7. Solve.

$$\ln(x+3) + \ln(x-2) = \ln 14$$

Answer: _____

8. Use the change of base formula to evaluate.

$$\log_7 50$$

Answer: _____

9. Compound interest: $5000 invested at 4% compounded continuously. Find the amount after 10 years.

$$A = Pe^{rt}, \quad P = 5000,\ r = 0.04,\ t = 10$$

Answer: _____

10. A population starts at 1000 and doubles every 5 years. Write the exponential model and find the population after 15 years.

$$P(t) = 1000 \cdot 2^{t/5}$$

Answer: _____

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Problems 2–3: expand vs condense — students confuse these directions. Problem 7 (log equation): extraneous root x=–5 must be rejected. Problems 9–10: applied context.

Solutions

1. Evaluate the logarithm.

$\log_2 32$

→ *$\log_2 32 = x$ means $2^x = 32 = 2^5$.*
→ *Answer: 5.*

Answer: 5

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2. Expand using logarithm properties.

$\log_b\left(\frac{x^3\sqrt{y}}{z^2}\right)$

→ *$\log(A/B)=\log A-\log B$. $\log(AB)=\log A+\log B$. $\log(A^n)=n\log A$.*
→ *$= \log x^3 + \log y^{1/2} - \log z^2 = 3\log x + \frac{1}{2}\log y - 2\log z$.*

Answer: $3\log_b x + \frac{1}{2}\log_b y - 2\log_b z$

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3. Condense into a single logarithm.

$2\ln x + \ln(x+1) - \ln 5$

→ *$2\ln x = \ln x^2$. Combine using product/quotient rules.*
→ *$\ln x^2 + \ln(x+1) - \ln 5 = \ln[x^2(x+1)/5]$.*

Answer: $\ln\left(\frac{x^2(x+1)}{5}\right)$

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4. Solve the exponential equation.

$3^{x+1} = 81$

→ *$81=3^4$. $3^{x+1}=3^4 \rightarrow x+1=4 \rightarrow x=3$.*

Answer: $x = 3$

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5. Solve the exponential equation using logarithms.

$5^x = 20$

→ *Take ln of both sides: $x\ln 5 = \ln 20$.*
→ *$x = \ln 20/\ln 5 = \log_5 20 \approx 1.861$.*

Answer: $x = \frac{\ln 20}{\ln 5} \approx 1.861$

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6. Solve the logarithmic equation.

$$\log_3(2x-1)=4$$

→ *Convert to exponential: $3^4=2x-1$.*
→ *$81=2x-1 \rightarrow 2x=82 \rightarrow x=41$.*

Answer: $x=41$

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7. Solve.

$$\ln(x+3)+\ln(x-2)=\ln 14$$

→ *$\ln[(x+3)(x-2)]=\ln 14 \rightarrow (x+3)(x-2)=14$.*
→ *$x^2+x-6=14 \rightarrow x^2+x-20=0 \rightarrow (x+5)(x-4)=0$.*
→ *$x=4$ (reject $x=-5$ since $x-2=-7$<0 makes ln undefined).*

Answer: $x=4$

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8. Use the change of base formula to evaluate.

$$\log_7 50$$

→ *Change of base: $\log_7 50 = \ln 50/\ln 7 = \log 10(50)/\log 10(7)$.*
→ *$\approx 3.912/1.946 \approx 2.012$.*

Answer: $\dfrac{\ln 50}{\ln 7}\approx 2.012$

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9. Compound interest: $5000 invested at 4% compounded continuously. Find the amount after 10 years.

$$A=Pe^{rt},\quad P=5000,\ r=0.04,\ t=10$$

→ *$A = 5000\cdot e^{(0.04\cdot 10)} = 5000\cdot e^{0.4}$.*
→ *$e^{0.4} \approx 1.4918$. $A \approx 5000\cdot 1.4918 \approx \$7459$.*

Answer: $A=5000e^{0.4}\approx \$7459$

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10. A population starts at 1000 and doubles every 5 years. Write the exponential model and find the population after 15 years.

$$P(t)=1000\cdot 2^{t/5}$$

→ *Doubling time 5 yrs → $P(t)=1000\cdot 2^{(t/5)}$.*
→ *$P(15)=1000\cdot 2^{(15/5)}=1000\cdot 2^3=1000\cdot 8=8000$.*

Answer: $P(15)=1000\cdot 2^3=8000$